

Data Science and Simulations – Assignment overview 2026

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Assignment 1: The compaction of intrinsically disordered proteins studied by machine learning

Proteins contain intrinsically disordered regions (IDRs) which do not fold into stable structures, but instead behave like flexible polymers. The conformation of IDRs is characterized by their level of compaction, which is dictated by specific physical and chemical attributes of the amino acid sequence. In this assignment, using a recently published large simulation dataset of IDRs, we will predict the level of compaction of IDRs from amino acid sequence features. To this end, guided by the solubility example studied throughout the course, we will implement a simple linear machine learning model and various deep neural network dense architectures.

Assignment 2: MD simulation of a protein with different force fields

As we learned through out the course, an empirical potential energy function, called force-field, is used to describe the interactions between atoms in molecular dynamics (MD) simulations. In the MD practical of the small B1 domain of protein G, we chose one of the conventionally used force-fields but we could have chosen a different one. In this assignment we will compare the force field we used originally for the practical (OPLS-AA/L all-atom for the protein and the SPC water model) with other force fields, one grouping few atoms into particles, named GROMOS, and another derived using machine learning, called GRAPPA.

Assignment 3: Correlation KO and compound inhibition in HTS experiments

During the course we have studied the effect of genetic knock-outs and their sensitivity in two cell lines in image-based screening assay and how this related to the gene expression profile differences. The specificity of genetic knock-out perturbations using CRISPR-Cas9 is relatively high if one uses guide RNA sequences with a low off-target score. However, theoretically only 66% of cells transfected with a guide RNA sequence will have an out-of-frame insertion or deletion, likely to disturb the open reading frame of the gene. Furthermore, in such experiments, not every cell automatically can be considered to be transfected with the guide RNA sequence. As an alternative, compounds targeting specific pathways or proteins can be used that are considered rather to affect all cells in the well (theoretically). However, a downside of compounds is, that they can be often a-specific and have a large variety of targets. Both approaches have their plus and minuses and in this project you will investigate the correlation between the responses of the cell lines to genetic perturbations with CRISPR compared to compound perturbations.

Assignment 4: Source publicly available data to interpret HTS results

During the course we have studied the effect of genetic knock-outs in an image-based screen in two cell lines and how this related to the gene expression profile differences. Although the data originated from two different papers, already indicating that publications are a rich resource for data to source, we have not looked in the context of other databases. There are a variety of databases out there that we can use to example for example the mutation rate of certain genes in cancer (<https://www.sanger.ac.uk/tool/cosmic/>), the dependency of cell lines on certain genes (<https://depmap.org/portal/>), or for example the single cell expression atlas (<https://www.ebi.ac.uk/gxa/sc/home>). In this specific project, we are mostly interested in how the observations in U2OS and A549 cells relate to a more clinical context. For this purpose we will make use of the cancer genome atlas (<https://portal.gdc.cancer.gov/>). This database holds information on the expression of genes but also on their mutation rate, includes different cancer entities and more. You will compare U2OS data with osteosarcoma cancer data and A549 data with lung cancer data. The focus is on identifying whether hits from the course – translate to the clinical setting.